Xiang Li | Curriculum Vitae

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RESEARCH INTEREST

Artificial Intelligence Security, Explainable Artificial Intelligence, Adversarial Attack, Backdoor Attack

EDUCATION

Southeast University **Nanjing, CHN**
M.E. in Computer Technology *September 2021 - June 2024 (expected)*

- Rank top 20%
- Advisor: Kai Dong

Nanjing University of Information Science and Technology **Nanjing, CHN**
B.E. in Computer Science and Technology *September 2017 - June 2021*

- Rank top 17%
- Advisor: Xiaolong Xu

RESEARCH EXPERIENCE

Adversarial Robustness of Self-explaining Models *2021 - 2023*

- It proposes the first systematical study on the adversarial robustness of self-explaining models. It performs the state-of-the-art adversarial attack for the scenario without human intervention, and proposes a new adversarial attack for the scenario with human intervention, to comprehensively evaluate the robustness of self-explaining models. The finding that the robustness does not always benefit from interpretability, but depends on the specific design in model architecture, challenges the common consensus that explainable models are often more robust.

Advanced Foundations of Backdoor Detection *2021 – 2023*

- It introduces a foundation beyond trigger identification, of detecting backdoors. It is based on two fundamental features of an infected model that emerge with backdoor injection. It can be applied to both black- and white-box models, and all types of backdoor attacks.
- It proposes FBD (a fast backdoor detection method for black-box models) which can be finished within 0.2 seconds and achieves better performance than existing methods, and TBD (a trusted backdoor detection method for white-box models) which can be finished within 1 minute, and achieves perfect detection precision and recall.

ACADEMIC PAPERS

Postgraduate Stage

- **On the Robustness of Self-Explaining Models**
  Kai Dong, Xiang Li, Zhen Ling, Ming Yang, Junzhou Luo
  [**WWW** submission] (**CCS** review scores: 532->4322)
- **Advanced Foundations of Backdoor Detection**
  Kai Dong, Xiang Li, Qianru Li, Chihao Jiang, Haodi Wang, Zhen Ling, Ming Yang, Xinwen Fu
  (**NIPS** review scores: 44433)

Undergraduate Stage

- **Service Offloading Oriented Edge Server Placement in Smart Farming**
  Jie Zhang, Xiang Li, Xuyun Zhang, Yuan Xue, Gautam Srivastava, Wanchun Dou
  Software: Practice and Experience (*SPE*).
- **Efficient Computation Offloading for Internet of Vehicles in Edge Computing-assisted 5G Networks**
  Shaohua Wan, Xiang Li, Yuan Xue, Wenmin Lin, Xiaolong Xu
  The Journal of Supercomputing (*TJSC*).

AWARDS & HONORS

- **The Second Price Scholarship**, SEU, 2022
- **The Third Price Scholarship**, SEU, 2021
- **Outstanding Dissertation**, NUIST, 2021
- **Outstanding Student Cadres**, NUIST, 2018-2021
- **The First Price Scholarship**, NUIST, 2018, 2020, 2021
- **The Second Price Scholarship**, NUIST, 2019

REFERENCES

- **Dr. Kai Dong**
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